



Novel layered triple hydroxide sphere CO₂ adsorbent supported copper nanocluster catalyst for efficient methanol synthesis via CO₂ hydrogenation

Jian Li, Tao Du^{*}, Yingnan Li, He Jia, Yisong Wang, Yanli Song, Xin Fang^{*}

SEP Key Laboratory of Eco-Industry, Northeastern University, Shenyang 110819, China

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ABSTRACT

Copper-based catalysts are playing important roles in CO₂ hydrogenation to methanol. Active copper provides sites for H₂ dissociation, whereas CO₂ adsorption and activation cannot be ignored either. Accordingly, novel Cu/LTH catalysts with copper nanoclusters supported on layered triple hydroxide (LTH) sphere CO₂ adsorbent were developed to improve microstructures and CO₂ adsorption capabilities simultaneously. The results show increases in copper content enlarge crystallite sizes yet reduce surface areas and pore volumes. All Cu/LTH catalysts consist of uniformly dispersed copper nanoclusters on external surfaces of LTH. Catalytic tests reveal remarkable properties of Cu/LTH catalysts compared with conventional catalysts. Further mechanism analyses demonstrate LTH can boost CO₂ adsorption capability at reaction conditions. Due to metal-support interactions, copper species in Cu₃/LTH are more easily reduced mainly in the form of Cu⁺, benefiting directional CO₂ conversion to methanol. Thus, it is feasible and efficient to ameliorate catalyst by incorporating adsorbent for enhanced CO₂ adsorption process.

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1. Introduction

The environment is one of the mostly concerned issues worldwide today. As the CO₂ concentration in atmosphere increases continuously due to excessive CO₂ emissions, the global temperature hits record highs in the past several years, leading to staggering rises of extreme weather events [1–3]. Consequently, the CO₂ utilization via catalytic conversion has aroused great attentions in academia and industry to reduce CO₂ emission and use CO₂ as the carbon resource for chemicals [4–6]. The methanol, an important basic chemical as well as a potential feedstock for many other value-added products (by methanol to olefin (MTO), methanol to propylene (MTP), etc.), is especially attractive among present CO₂ utilization technics [7–10].

Promising as the methanol synthesis via CO₂ hydrogenation is, its industrial application still suffers from many restrictions [11]. One of them is the thermodynamic limitation. As several competitive reactions including methanol synthesis from CO₂, methanol synthesis from CO and reverse water gas shift (RWGS) take place together, low-temperatures and high-pressures are favorable according to Le Chatelier's principle. One problem lies in poor cat-

alytic activities of conventional catalysts at low temperatures, which usually can hardly exhibit considerable activities until ~250 °C [12]. In order to maintain high methanol selectivity, the reaction pressure has to be elevated, and thus there comes the other problem of substantially increased operation costs [13]. According to the reported technical and economic assessments, costs of compressors for the system pressure increase account for 45 % of the total equipment costs, and energy consumptions during the compressing process account for 66 % of the total electricity costs [14]. Therefore, it's significant to improve properties of catalysts before they can be applied in industry.

A large number of pioneering research works has been carried out for that target. Commonly used methods include component modification and microstructure regulation [15,16]. Briefly, transition metal elements and/or precious metal elements such as Zn, Zr, Ce, Ti, Pt, etc. can be incorporated in catalyst systems to fine-tune metal dispersions and metal-oxide interfaces to promote the catalytic activity, product selectivity and/or performance stability [6,17–22]. On the other hand, catalyst precursors with specific structures are also adopted to regulate microstructures of catalyst directly, in which metal-organic frameworks (MOFs) and layered double hydroxides (LDHs) are verified as ideal candidates [23–27]. Moreover, many supports including metal oxides and porous materials have attracted great attentions of researchers to generate novel hybrid catalysts with high surface areas and metal disper-

^{*} Corresponding authors.

E-mail addresses: dut@mail.neu.edu.cn (T. Du), fangxin@mail.neu.edu.cn (X. Fang).

sions. Observed metal-support interactions (MSIs) in these catalysts also provide possibilities and feasibilities for remarkable properties in reactions [28–30]. Apart from common supports like ZnO, active alumina, porous silica, carbon nanotube/nanowire, etc., the LDH has been confirmed in our previous works as an interesting support for Cu-ZnO-ZrO₂ nanoparticles to facilitate copper dispersion and CO₂ adsorption [24], and the novel Mg-Zn-Al type layered triple hydroxide (LTH) developed in our subsequently experiments attracts our special attentions. On the one hand, it shows distinctive hollow sphere morphology with rich external surfaces to accommodate nanoclusters. On the other hand, the alkaline metal oxide MgO in LTH, as it helps increase the number of CO₂ and H₂ adsorption active sites, can enrich basic sites and strengthen metal-support interactions [31–34]. They both have been confirmed beneficial for catalytic performance improvements, and hence the LTH is expected as a promising support for copper nanoclusters to improve surface area, metal dispersion, high-temperature CO₂ adsorption and thus catalytic performance. However, relative study is absent so far.

In this work, therefore, a series of novel LTH sphere supported copper cluster catalysts (viz., Cu/LTH) were prepared with different copper contents. The samples were characterized to draw clear pictures about their textural properties, and then tested in catalytic reaction to understand their performances in CO₂ hydrogenation process. In further analyses in mechanism, the optimal Cu/LTH catalyst, conventional Cu-ZnO-Al₂O₃ (CZA) catalyst and reference SBA-15 supported copper cluster (Cu/SBA-15) catalyst were compared in aspects of high-temperature CO₂ adsorption capacity, reductivity and metal status, and impacts of LTH were discussed in detail.

2. Materials and methods

2.1. Reagents

Copper nitrate (Cu(NO₃)₂·3H₂O), zinc nitrate (Zn(NO₃)₂·6H₂O), magnesium nitrate (Mg(NO₃)₂·6H₂O), aluminum nitrate (Al(NO₃)₃·9H₂O), urea (CH₄N₂O), polyvinyl pyrrolidone (PVP) and ethanol (EtOH) were supplied by the Sinopharm. The ultra-pure water was home-made (CANSI). All reagents were used directly without further purification.

2.2. Preparation of catalysts

The catalysts were prepared by impregnating copper nanoclusters onto porous LTH spheres. Herein, the Mg-Zn-Al type LTH was selected as the support (refer to the SI for detailed preparation method). In a typical procedure, 2 g of activated LTH powders and a certain amount of PVP (with a PVP: Cu(NO₃)₂·3H₂O mass ratio of 1:10) were uniformly dispersed in 25 mL of ethanol to obtain the mixture (viz., Sol. 1). Then, 25 mL ethanol solution of Cu(NO₃)₂ (viz., Sol. 2) was added in the Sol. 1 and stirred for 6 h at the ambient temperature. Afterwards, the precipitate was centrifuged, washed with ethanol for 3 times, dried in an oven at 80 °C overnight and calcined at 400 °C for 4 h to obtain the catalyst. Since different Cu(NO₃)₂ concentrations ranging from 0.06 mol/L to 0.25 mol/L were adopted in Sol. 2 to guarantee different copper contents in catalysts, the as-prepared catalysts were labeled as Cu1/LTH, Cu2/LTH, Cu3/LTH and Cu4/LTH, respectively.

Additionally, the benchmark Cu-ZnO-Al₂O₃ (CZA) catalyst was prepared by conventional co-precipitation method. Another SBA-15 supported copper cluster (Cu/SBA-15) catalyst was also prepared by wet impregnation method as the reference, in which either high-temperature CO₂ adsorption or strong MSIs can be neglected. Detailed preparation procedures were listed in the SI.

2.3. Catalyst characterization

Chemical compositions of prepared samples were quantitatively analyzed by the X-ray fluorescence (XRF; ZSXPrimus II, Rigaku). Crystalline phases of all sample were identified using the X-ray diffraction (XRD; D8 Advance Diffractometer, Bruker) with a Cu K α radiation. XRD patterns were recorded for 2 θ values from 20° to 80°. Crystallite sizes of samples were further calculated using the Scherrer Equation. Fourier transform infrared spectroscopy (FT-IR; Cary 660, Agilent) were measured to characterize types of chemical bonds within the wavenumber of 2000 to 400 cm⁻¹. Morphologies of samples were presented by the scanning electron microscope (SEM; Sigma 300, Zeiss). All samples were coated with a thin layer of aurum prior to measurement for better electroconductibility. Low-temperature N₂ adsorption-desorption isotherms were tested at -196 °C (ASAP 2460, Micromeritics). The Brunauer-Emmett-Teller (BET) surface areas were calculated accordingly. Ex-situ X-ray photoelectron spectroscopy (XPS) analyses were performed to analyze the valence of elements on a ThermoFischer ESCALAB 250Xi spectrometer using Al K α (h ν = 1486.6 eV) radiation. The sample was reduced by H₂ prior to analyses, and the C1 s (284.8 eV) was used as reference for the calibration of binding energies. Hydrogen temperature programmed reduction (H₂-TPR) was conducted on AutoChem1 II 2920 equipped with a thermal conductivity detector (TCD). The sample (about 150 mg) was pretreated under flowing argon for 2 h at 300 °C to remove moisture and absorbed water. The sample was then cooled down to 50 °C in Ar flow. Finally, the sample was heated from 50 °C to 600 °C (with a ramp rate of 5 °C/min) in 10 % H₂ flow (balanced with Ar), during which the H₂ consumption was measured by the TCD.

2.4. Catalytic property analysis

Catalytic reaction experiments were conducted in a fixed-bed microreactor (column ID 10 mm $\times$ L 500 mm). The reactor included three sections: (i) gas supply section, (ii) reaction section, and (iii) analysis section. In the gas supply section, three mass flow controllers were used to maintain the feed gas flowrate, where N₂ was used to purge the whole system, H₂ was used for in-situ reduction and H₂/CO₂ pre-mixed gas (H₂: CO₂ = 3:1) was used as the feedstock. In the reaction section, the maximum reaction temperature could be controlled to 400 °C and the reaction pressure could be maintained by a back pressure regulator. In the analysis section, the outlet gas was analyzed by the digital bubble flowmeter (DBF) and online gas chromatograph (GC; GC-102, HENVEN) to measure flowrate and compositions, respectively. The GC was equipped with thermal conductivity detector (TCD) and hydrogen flame ionization detector (FID).

In a typical test, 1 g of tableted sample (60–80 mesh) and 1 g of quartz sand (50–70 mesh, Sigma-Aldrich) were mixed and filled in the column. It was in-situ reduced in H₂ at 300 °C for 6 h and cooled in N₂ flow. Then, the sample was heated and maintained at the set temperature and pressure in H₂/CO₂ pre-mixed gas (with GHSV of $\sim$ 3000 h⁻¹). After 4 h, the reaction reached a steady state, and composition and flowrate of the outlet gas were measured every 2 h. As no carbon-containing product apart from methanol and CO could be observed, the carbon balance was established based on CO₂, methanol and CO, and confirmed with low error within 100 $\pm$ 2% to give reliable results.

The CO₂ conversion (X_{CO_2}) were calculated as.

$$X_{\text{CO}_2} = \frac{f_{\text{in}} \times P_{\text{CO}_2, \text{in}} - f_{\text{out}} \times P_{\text{CO}_2, \text{out}}}{f_{\text{in}} \times P_{\text{CO}_2, \text{in}}} \times 100\%$$

where f_{in} (NmL·min⁻¹) was the inlet flowrate, $p_{CO_2,in}$ (%) was the molar fraction of CO₂ in the inlet, f_{out} (NmL·min⁻¹) was the outlet flowrate and $p_{CO_2,out}$ (%) was the molar fraction of CO₂ in the outlet.

The selectivity of methanol and CO (S_{MeOH} and S_{CO}) were calculated as.

$$S_{MeOH} = \frac{f_{out} \times p_{MeOH,out}}{f_{in} \times p_{CO_2,in} - f_{out} \times p_{CO_2,out}} \times 100\%$$

$$S_{CO} = \frac{f_{out} \times p_{CO,out}}{f_{in} \times p_{CO_2,in} - f_{out} \times p_{CO_2,out}} \times 100\%$$

where $p_{MeOH,out}$ and $p_{CO,out}$ (%) were the molar fraction of methanol and CO in the outlet, respectively.

The space time yield (STY) of product i was calculated as.

$$STY_i = \frac{f_{in} \times p_{CO_2,in} \times X_{CO_2} \times S_i}{W_{cat.} \times t} \times 100\%$$

where i is the product viz., MeOH or CO, and $W_{cat.}$ is the weight of catalyst (g), and t was the reaction time (h).

3. Results and discussion

3.1. Textural properties of catalysts

3.1.1. Copper loading rates and pore structures

As different initial copper concentrations in ethanol solution are used, the samples show different copper loading rates. Measured copper contents in Cu(1-4)/LTH by XRF are 10.3, 13.7, 18.1 and 24.8 wt%, respectively. They are basically consistent with theoretical copper loading rates calculated according to original Cu:LTH mass ratios, and hence the process of copper nanocluster loading onto LTH is successful and effective.

The copper content impacts pore structures directly. The BET specific surface area of LTH is 270.9 m²/g, which provides a lot of loading sites for copper. As can be observed in Fig. 1(a), this value drops from 237.6 m²/g in Cu1/LTH to 175.9 m²/g in Cu4/LTH when the copper content increases. Still, they are times higher than conventional Cu based catalysts (usually under 80 m²/g) [17,20,35,36]. In the aspect of pore volume, the situation is similar. Their pore volume gradually decreases from 0.35 cm³/g to 0.24 cm³/g with the increase of copper content. This is because that when there are more copper in samples, they tend to cover more outer surfaces and block more pores of LTH, leading to declined BET surface area and pore volume. Additionally, low-temperature N₂ adsorption-desorption isotherms of samples shown in Fig. 1(b) are typical IV

type accompanied with obvious H1 hysteresis loops, indicating the presence of complex pore structures associated from LTH and accumulations of copper nanoclusters [37].

3.1.2. Crystal structures

Fig. 2 presents XRD patterns of Cu(1-4)/LTH samples with different copper contents. Basically, Cu(1-4)/LTH samples consist of CuO and LTH in crystal. The CuO, which can be reduced and plays crucial roles in CO₂ hydrogenation [6,15,16,38,39], can be detected by identifying the corresponding diffraction peaks at 35.5, 38.7, 48.7°, etc. (JCPDS 45-0937), demonstrating that copper is successfully loaded on LTH and exists in the form of metal oxide. The peaks are clear and sharp, so that CuO species in samples are well crystallized. Moreover, intensities of these diffraction peaks increase along with the copper content, and that reflects increasing amounts of CuO. Apart from diffraction peaks of CuO, Cu(1-4)/LTH samples also show peaks at 31.8, 43.6 and 62.8° in their XRD patterns, which are identical with the LTH in Fig. 2. It demonstrates well retained LTH structures in Cu(1-4)/LTH samples, and the weakened intensities of these peaks result from reduced amounts of LTH.

Subsequent analyses based on XRD results by the Scherrer equation show that the crystallite size of CuO increases continuously with raised copper content from 13.8 nm (Cu1/LTH) to 20.5 nm (Cu4/LTH) according to Fig. 1(a). It indicates that large CuO may be formed when the copper content is high, which is not favorable for copper dispersion. To achieve ideal catalytic performance, certain amounts of copper are required to provide necessary active sites (Cu⁰), but high copper dispersions are also admired to guarantee efficiently usages of copper. Therefore, an optimal copper content needs to be determined to reach a balance between copper content and CuO size.

3.1.3. Chemical bonds

Chemical bonds in Cu(1-4)/LTH samples are further analyzed by FTIR to identify the status of LTH supports. As can be observed in Fig. 3, peak shapes of the samples are similar to that of LTH, indicating that there are well retained LTH in as-prepared catalysts. Shapes and positions of their absorption peaks of Cu(1-4)/LTH samples are also very similar, which means that they have identical types of chemical bonds. Specifically, the peak found at ~1640 cm⁻¹ suggests the bending vibration of H₂O [40]. Stretching vibration and anti-stretching vibration of NO₃⁻ can be observed at ~1050 cm⁻¹ and ~1380 cm⁻¹ associated from the interlayer anion of NO₃⁻ in samples [23,41]. Besides, there are several absorption peaks appearing between 400 and 800 cm⁻¹ [42], which are due

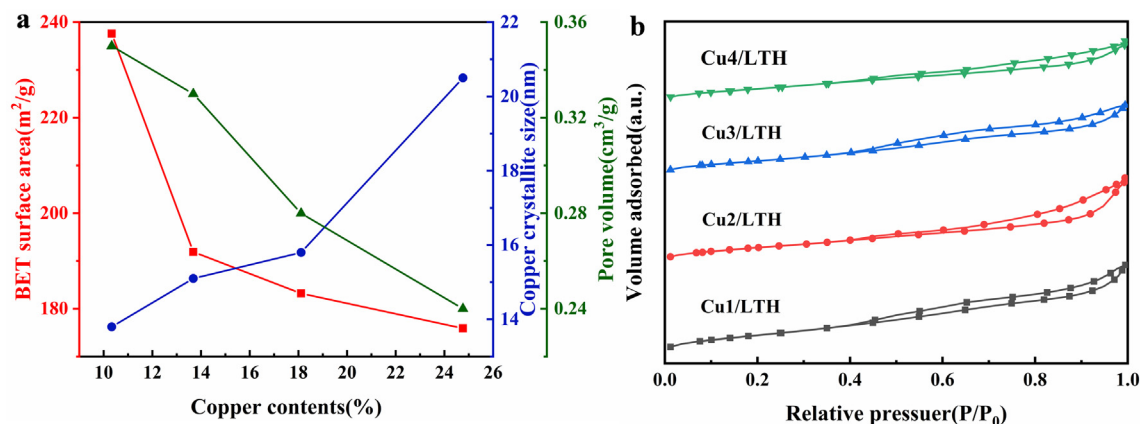


Fig. 1. (a) BET surface area, copper particle diameter and pore volume and (b) Low-temperature N₂ adsorption isotherm of as-prepared catalysts with different copper contents.

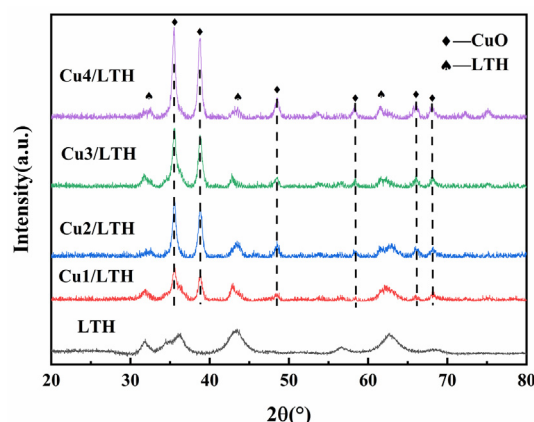


Fig. 2. XRD patterns of Cu(1–4)/LTH and LTH.

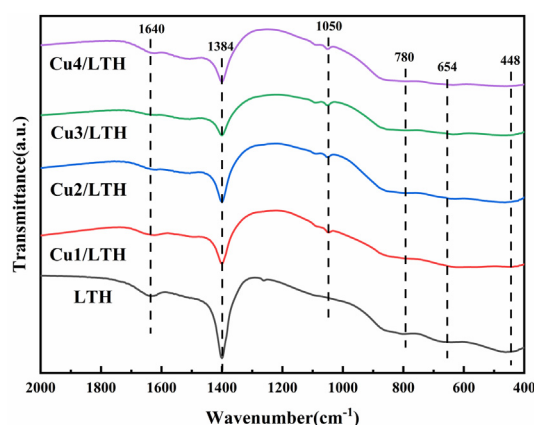


Fig. 3. FTIR spectra of Cu(1–4)/LTH and LTH.

to the combined effect of various metal ions and oxygen (viz., M–O bonds) [43].

3.1.4. Morphologies

Morphologies of LTH and Cu(1–4)/LTH samples in Fig. 4 reflect their differences in copper content. Primarily, they all perfectly inherit the morphology of activated LTH with a sphere appearance, and copper nanoclusters are supported on external surfaces. Due to raised copper contents in samples, amounts of nanoclusters also increase from Cu1/LTH (Fig. 4(a)) to Cu3/LTH (Fig. 4(c)). However, when the copper content reaches a high level of 24.8 wt%, i.e., Cu4/LTH shown in Fig. 4(d), there are too many copper nanoclusters on surfaces to form particle agglomeration, which may have an adverse effect on the CO₂ hydrogenation performance.

Based on characterizations above, the Cu3/LTH catalyst shows the optimal textural property among prepared samples.

3.2. Catalytic properties

3.2.1. Impact of copper content on catalytic performance

Catalytic properties of Cu(1–4)/LTH samples with different copper amounts was investigated under the typical reaction condition of 250 °C and 3.0 MPa for CO₂ hydrogenation to methanol. Primarily, the LTH is proven with no activity for CO₂ hydrogenation to methanol (see SI), and the results of Cu(1–4)/LTH samples are shown in Fig. 5. As the copper content increases from ~10 % to ~25 %, the CO₂ conversion of the samples increases from 12.2 % to 23.0 % first and then decreases to 21.9 %, while the methanol

selectivity of the catalysts has little differences, all between 73 % and 75 %. Therefore, as shown in Fig. 5(b), the space time yield of methanol generally shows a trend of increasing first and then decreasing with that of Cu3/LTH reaching the highest. Specifically, the STY_{MeOH} of Cu3/LTH reaches 145.2 mg_{MeOH}·g_{catalyst}^{−1}·h^{−1}, which is about 1.85 times that of Cu1/LTH and 1.05 times that of Cu4/LTH. This phenomenon occurs because the copper, which is the active component of the reaction, can provides more reaction active sites with increasing loading rate. But when the loading rate is too high, it will lead to the overgrowth of copper species particles and obvious agglomeration as observed in Table 1 and Fig. 4, thus damage the overall catalytic performance. Based on the analysis above, the best copper loading for LTH is 20 %.

3.2.2. Property stability

Sample Cu3/LTH has shown favorable catalytic performance in previous section, and its property stability is also of great importance. Fig. 6 presents its property changes along with the reaction time in continuous reactions at 250 °C, 3.0 MPa and ~3000 h^{−1} (GHSV). Within 4 to 10 h from the beginning of the reaction, the CO₂ conversion was 23.0 ± 1.0 % and the methanol 74.5 ± 1.5 %. And the CO₂ conversion was 23.5 ± 1.5 % and the methanol 74.0 ± 2.0 % after 20 h of reaction to 32 h. So, it can be seen that the CO₂ conversion and methanol selectivity show little change with the progress of the catalytic reaction and there is no deactivation after 32 h of continuous reaction. Thus, the Cu3/LTH shows considerable stability.

3.2.3. Comparison of catalytic properties among Cu3/LTH, CZA and Cu/SBA-15

Table 1 presents the catalytic properties of as-obtained samples and some reported catalysts. It can be seen that under the condition of 250 °C and 3 MPa, the CO₂ conversion and methanol selectivity of Cu3/LTH are significantly better than those of conventional as-obtained CZA catalysts (15.4 % and 57.5 %, respectively, similar to the reported CZA [44]) and 0.9 times higher in the space time yield of methanol. Compared with Cu/SBA-15, the catalytic properties of Cu3/LTH are obviously superior in catalytic properties and about 13.5 times higher in the space time yield of methanol. If the methanol STY based on unit mass of Cu (STY_{MeOH,Cu}) is taken in consideration, suppositories of Cu3/LTH are rather significant, which is 3.8 and 12.4 times higher than CZA and Cu/SBA15, respectively.

Besides, the Cu3/LTH shows superior catalytic properties in comparison to the novel Cu-based catalysts loaded on other supports. For instance, Cu/ZnO/ZrO₂/SBA-15 shows the equivalent CO₂ conversion with Cu3/LTH, while the methanol selectivity is only 41.6 % of Cu3/LTH. Except for Cu based catalysts, there are many Cu-free CO₂ hydrogenation catalysts with other types of active sites like Pd, etc. Although they have relatively high methanol selectivity, their catalytic performance is limited by the low CO₂ conversion, usually less than 20 %. In summary, Cu3/LTH exhibits a competitive edge compared with other catalysts.

3.3. Metal-support interactions and impacts on catalytic performance

3.3.1. Impacts of LTH on CO₂ adsorption capacity

Adsorption of CO₂ are acknowledged crucial during the CO₂ hydrogenation process because CO₂ molecules must be adsorbed and then activated (*CO₂) before they can be further converted, and thus considerable CO₂ adsorption capacities at high temperatures and pressures are indispensable. As we have explicated in our previous work [51], CO₂ hydrogenation catalysts adsorb CO₂ by physisorption, chemisorption and surface reactions. In this work, CO₂ adsorption capacities of several typical catalysts were measured at 250 °C and high pressures up to 1.2 MPa for the first

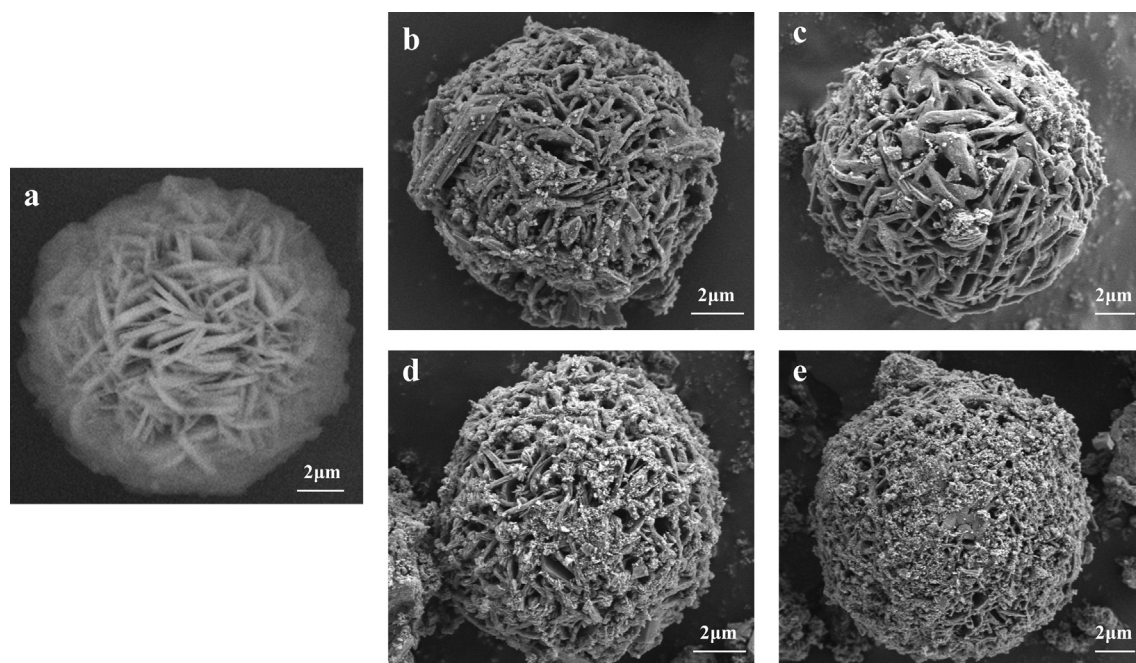


Fig. 4. SEM images of (a) LTH, (b) Cu1/LTH, (c) Cu2/LTH, (d) Cu3/LTH, (e) Cu4/LTH.

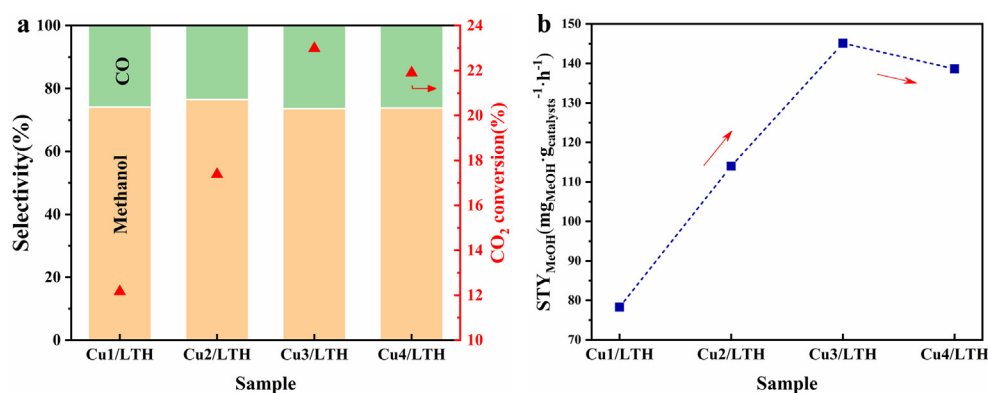


Fig. 5. Catalytic properties of Cu(1–4)/LTH samples with different copper amounts (reaction temperature: 250 °C, pressure: 3.0 MPa, GHSV: ~3000 h⁻¹).

Table 1

Catalytic properties of as-obtained samples and some reported catalysts.

Catalysts	T (°C)	P (Mpa)	GHSV (h ⁻¹)	X _{CO2} (%)	S _{MeOH} (%)	STY _{MeOH} (mg _{MeOH} ·g _{catalysts} ⁻¹ ·h ⁻¹)	Ref
Cu3/LTH	250	3.0	3000	23.0	73.6	145.2	This work
CZA	250	3.0	3000	15.4	57.5	76.3	
Cu/SBA-15	250	3.0	3000	7.7	15.1	10.0	
CZA-C-1.25	240	3.0	3600	16.2	63.8	130	[44]
Cu/ZrO ₂ /CNT-3	260	3.0	3600	16.3	43.5	83.8	[45]
CuZnAl-C-1.00	240	3.0	3600	14.6	63.6	120.2	[44]
CuNi ₂ /CeO ₂ -NT	260	3.0	6000	17.8	78.8	579.9	[46]
CuZnZr/CuB ₁₂	250	5.0	3000	10.9	97.1	103	[47]
Cu/ZnO/ZrO ₂ /SBA-15	250	3.0	44,000	19.2	30.6	376	[48]
1PdZn/CeO ₂	220	2.0	3600	14.1	97.2	165.9	[49]
Pd/ZnO/CNTs	250	3.0	1800	6.3	99.6	37.1	[50]

time. Fig. 7 shows that the benchmark CZA catalyst can adsorb CO₂ mainly because of presences of ZnO species by chemisorption and surface reactions [52]. When there is only Cu and no CO₂ adsorbent in Cu/SBA-15, CO₂ adsorption capacities are relatively low and they increase with the pressure due to accumulations of CO₂ molecules in SBA-15 channels, which is a typical physisorption. Situations are

totally changed by the involvement of LTH. The LTH presents dramatic capabilities to adsorb CO₂ at high temperature and pressure via physisorption, chemisorption and surface reactions because of its rich and suitable basic active sites. According to the supplementary CO₂-TPD profile of Cu3/LTH in Figure S1, rich weak (α peak), moderate (β peak), and strong (γ peak) basic sites can be observed,

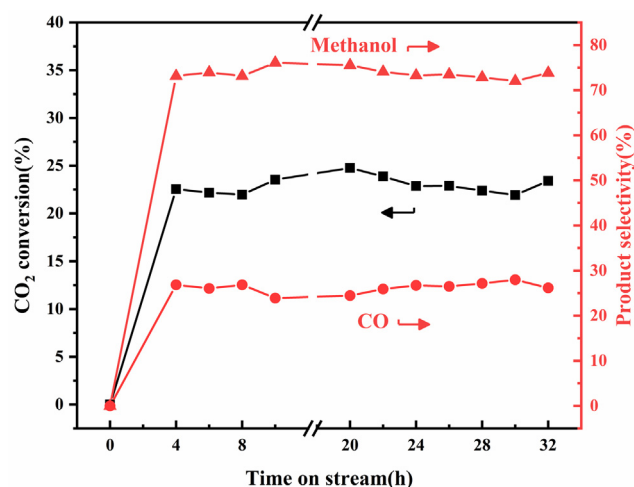


Fig. 6. Stability of Cu3/LTH at 250 °C and 3.0 MPa.

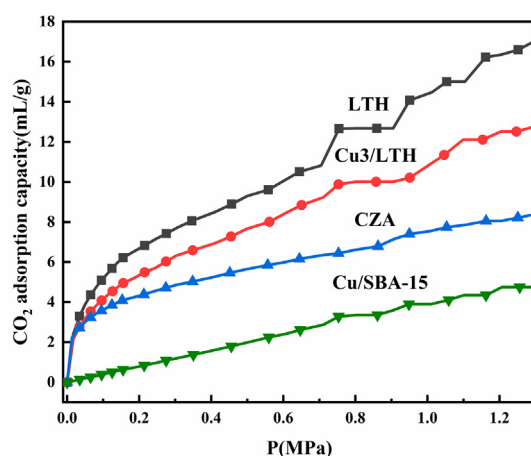


Fig. 7. Adsorption isotherms of CO₂ at 250 °C (LTH, Cu3/LTH, CZA and Cu/SBA-15).

revealing proper surface basicity for CO₂ adsorption and activation. Therefore, the hybrid Cu3/LTH catalyst possesses apparently higher CO₂ uptakes than the benchmark CZA catalyst. CO₂ molecules, alternatively, can be activated via different routes from conventional ones and be converted to methanol more directionally.

3.3.2. Impacts of LTH on reductivity

Statuses of Cu species in Cu3/LTH catalyst can be fine-tuned by MSIs between Cu and LTH as illustrated by H₂-TPR shown in Fig. 8. In the H₂-TPR profile of LTH, there is no H₂ consumption peak, indicating no component in LTH can be reduced within the investigated temperature range. In Cu/SBA-15 where the MSIs is too weak and can be ignored, though Cu species herein can be reduced, the peak is wide and flat, which means copper species distributes nonuniformly and some copper oxides are difficult to be reduced due to their large sizes. By comparison, results of CZA and Cu3/LTH are complex but interesting. Both profiles of Cu3/LTH and CZA can be deconvoluted into three Gaussian peaks, which are denoted as α , β and γ peak. The α peak is attributed to the reduction of highly dispersed CuO species, the β peak is attributed to the respective reduction of the CuO species interacted with supports and the γ peak was attributed to the bulk CuO species [20]. Specifically, peaks of Cu3/LTH are centered at 234, 266 and 293 °C, while the peaks of CZA are centered at 270, 324 and 351 °C. Thus, the reduction temperatures of Cu3/LTH shift to

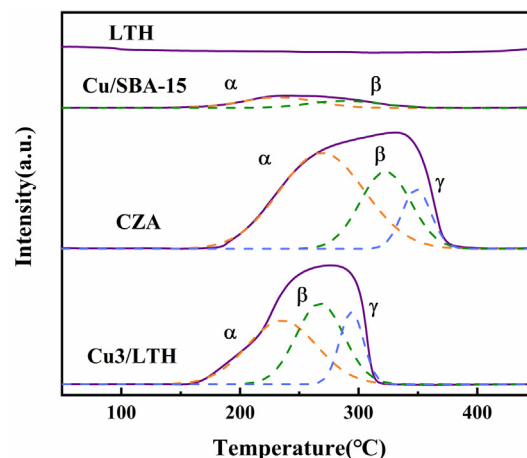


Fig. 8. H₂-TPR profiles of Cu3/LTH, CZA, Cu/SBA-15 and LTH.

low-temperature regions compared with CZA, and the CuO supported on LTH can be reduced under lower temperature than CZA. That reflects reduced CuO crystalline sizes and/or improved metal dispersions [53,54], which are all beneficial for efficient hydrogenation conversion of CO₂.

3.3.3. Impacts of LTH on element status

The element status can provide direct evidence of MSIs in the Cu3/LTH catalyst, and thus XPS tests are conducted using Cu3/LTH, benchmark CZA and reference Cu/SBA-15. The XPS spectrum surveyed from 0 to 1350 eV (Fig. 9(a)) indicates that the samples contain characteristic peaks of Cu, Zn, Al, Mg, C, O, Si and other elements. The C element comes from the CO₂ in the air adsorbed on the surface of the material, and the peak position is often used as a reference system to calibrate the peak position of other elements.

The status of Cu is of great importance for CO₂ hydrogenation. In Fig. 9(b), all the samples exhibit Cu 2p_{3/2} peak at ~932.2 eV and Cu 2p_{1/2} peak at ~953.6 eV with satellite peaks appearing at 962.2 eV and 940.0–950.0 eV. That suggests all samples contain copper in multiple valence states though they have been reduced. However, Cu⁰ and Cu⁺ cannot be well distinguished by Cu 2p spectrum due to their very close binding energies, and the X-ray induced Auger spectrum (XAES) of Cu LMM is hence adopted, in which the standard peaks of Cu⁰ and Cu⁺ are at approximately 918.8 and 916.4 eV, respectively. As can be observed in Fig. 9(c), there is obvious peak at 916.7 eV (corresponding to Cu⁺) for Cu3/LTH, suggesting that copper mainly exists in the form of Cu⁺. Different from Cu3/LTH, the kinetic energy of the peak of CZA is about 917.6 eV, which may be the result of the overlapping of the peaks at 916.4 eV and 918.8 eV. Although both Cu⁺ and Cu⁰ species are believed to the activity of Cu-based catalysts, a proper ratio of Cu⁺ and Cu⁰ is important for the catalytic performance in CO₂ hydrogenation to methanol [38,39,55]. In addition, the peak of Cu/SBA-15 is significantly lower compared with Cu3/LTH and CZA, indicating that the CuO supported on SBA-15 is difficult to be reduced, which can also explain the weak catalytic properties of Cu/SBA-15. Combined with the subsequent result of H₂-TPR, the reason for this phenomenon can be contributed to the aggregation of CuO which makes it difficult to be reduced [56].

The results of Zn 2p are shown in Fig. 9(d), and the results of Cu/SBA-15 is displayed as a line because it does not contain Zn. It is reported that ZnO can improve the dispersion of copper, and due to the interaction between Cu and ZnO, ZnO can also adjust the electronic properties [52,56–59].

The results of Mg 1s are shown in Fig. 9(e) and MgO only exists in Cu3/LTH. The addition of basic MgO decreases the surface den-

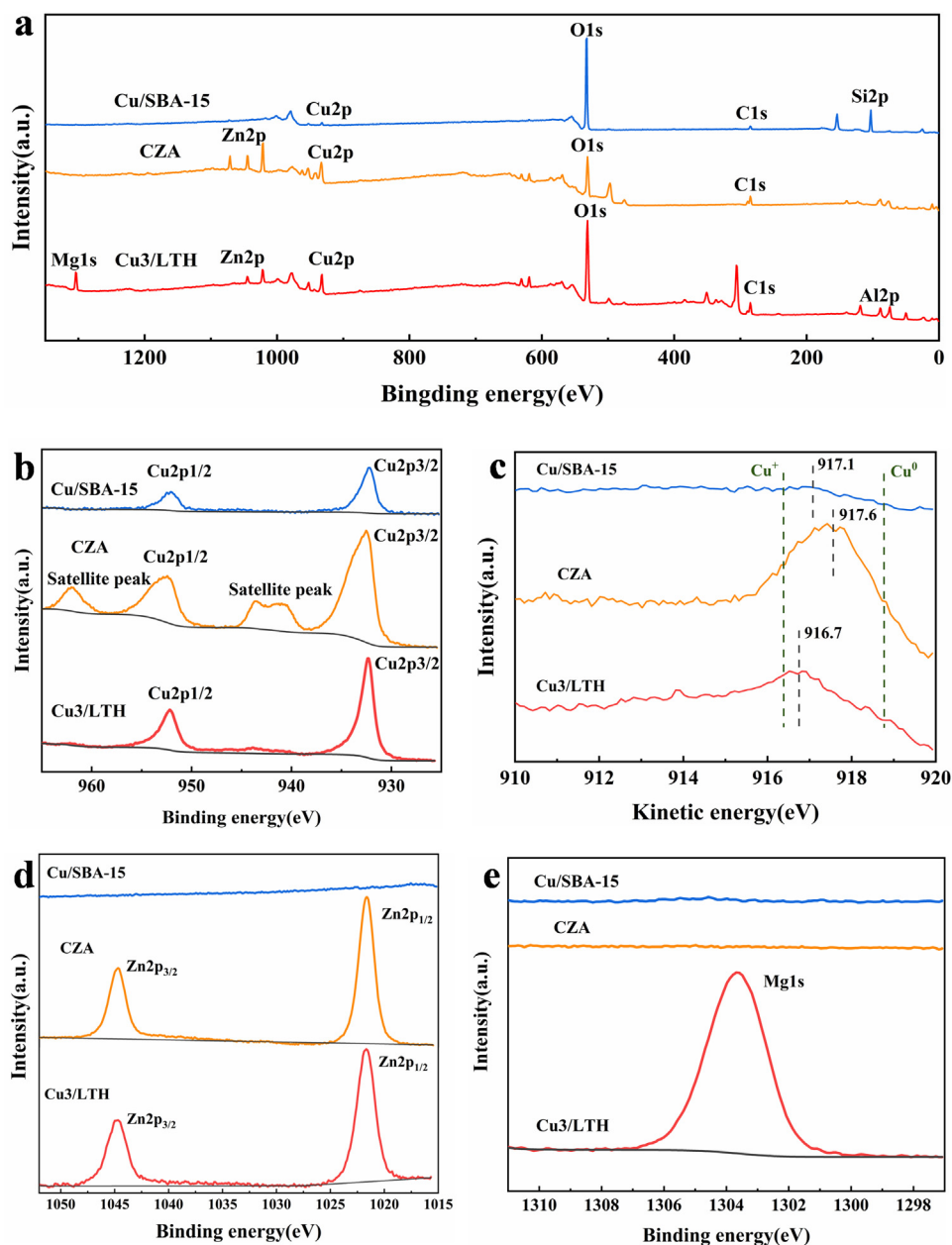


Fig. 9. (a) XPS survey spectra, (b) Cu 2p XPS spectra, (c) Cu XAES spectra, (d) Zn 2p XPS spectra and (e) Mg 1s XPS spectra peaks of Cu3/LTH, CZA and Cu/SBA-15 catalysts after reduction.

sity of Cu during synthesis and increases the dispersion of Cu species [60]. In addition, the MgO has been reported as a CO₂ activation center and beneficial for CO₂ hydrogenation [61].

Therefore, the CuO loaded on LTH is more likely to be reduced to Cu⁺ during the reduction, promoting the conversion of CO₂ to methanol. Meanwhile, the additions of ZnO can improve the dispersion of copper and adjust the electronic properties and MgO facilitates the activation of CO₂ in catalytic reactions. Thus, the LTH is significantly superior as a support in the reaction of CO₂ hydrogenation to methanol.

3.3.4. Proposed reaction mechanism

Based on the above results, the copper nanoclusters are supported on the surface of the spherical LTH with good dispersion (as shown in Fig. 10). As reported in previous theoretical works

of CO₂ hydrogenation to methanol, the adsorption and dissociation of H₂ molecules occur on the surface copper, and the adsorption and activation of CO₂ mainly take place on Zn species [52,62–64]. Herein, the adsorption and activation of CO₂ occur on LTH. Then, the atomic hydrogen on the surface of copper is removed by spillover to metal oxides and reacts with the carbon species, which are adsorbed on the surface of these oxides, to form methanol [65,66]. Compared with previously reported catalysts, in exception to reducing crystal size and/or improving metal dispersion, the addition of LTH can also improve the CO₂ adsorption capability of Cu/LTH at high temperatures and pressures, leading to the increase of CO₂ concentration on LTH surface. Since the active copper is tightly loaded on LTH, when the CO₂ concentration on the surface of LTH increases, it is equivalent to increasing the CO₂ partial pressure of the reaction system and enhancement of CO₂

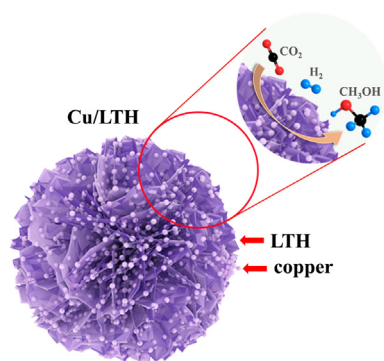


Fig. 10. Schematic of Cu/LTH catalyst.

activation on its surface. Moreover, copper species in Cu/LTH are easier to be reduced mainly in the form of Cu^+ due to the MSIs, which is beneficial for the conversion of CO_2 to methanol.

4. Conclusion

In this study, novel Cu/LTH catalysts, with copper nanoclusters supported on layered triple hydroxide (LTH) sphere CO_2 adsorbent, were prepared by impregnation to improve microstructures and CO_2 adsorption capabilities simultaneously, and the influence of copper loading was discussed. The results show that Cu/LTH catalysts present a unique hollow spherical shape and well dispersed copper nanoparticles are observed on the external surfaces of Cu/LTH without obvious agglomeration. And the Cu_3/LTH catalyst with 18.1 wt% copper loading presents the best catalytic property and consistent performance in continuous reactions. Further analyses in mechanism demonstrate that the addition of LTH can improve the CO_2 adsorption capability of Cu_3/LTH at high temperatures and pressures. And due to MSIs, copper species loaded on LTH is easier to be reduced in reaction and more likely to be reduced to Cu^+ , promoting the conversion of CO_2 to methanol and improving the catalytic properties accordingly, it is feasible and efficient to ameliorate catalyst by incorporating porous adsorbent for enhanced CO_2 adsorption process.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcat.2022.03.020>.

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